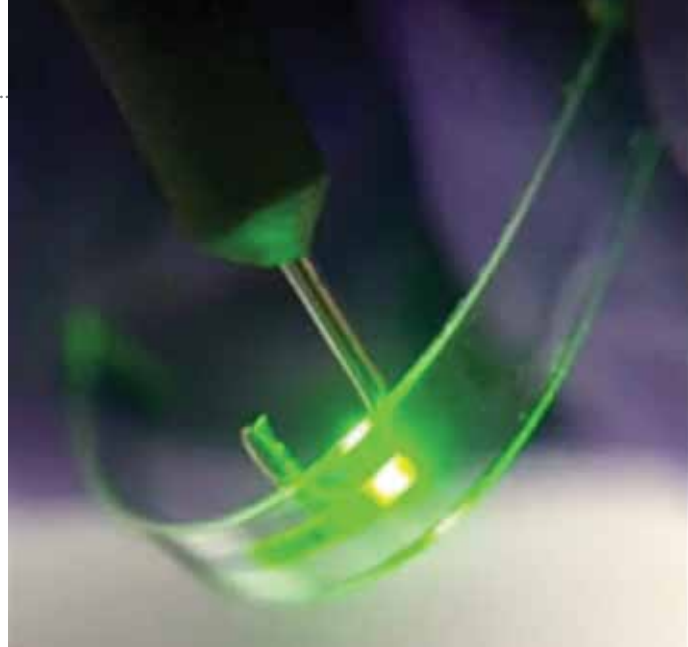


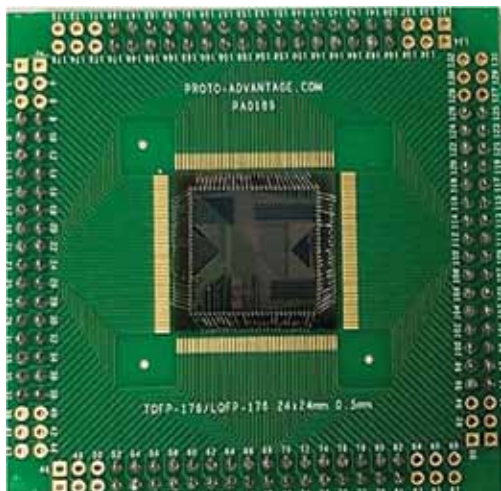
## Flexible LEDs created with inkjet printer

You could attach two screens with a hinge and call it foldable, but what if you could roll it up and put it in your wallet? Or stretch it around your wrist to wear it as a watch? This has now been made a reality by the researchers at the McKelvey School of Engineering, Washington University in St. Louis. They have developed a new material for creating LEDs and OLEDs using a novel way—an inkjet printer. Inkjet fabrication saves materials as it precisely deposits the printing material without any wastage. The process is much faster as well, cutting fabrication time from more than five hours to less than 25 minutes. Moreover, inkjet printing method has the potential to reshape the future of electronics for printing a variety of unconventional substrates, including those that would not lend themselves to stability while spinning such as rubber.



A flexible PeLED (Credit: <https://source.wustl.edu>)

## Carbon nanotubes to protect electronics in outer space

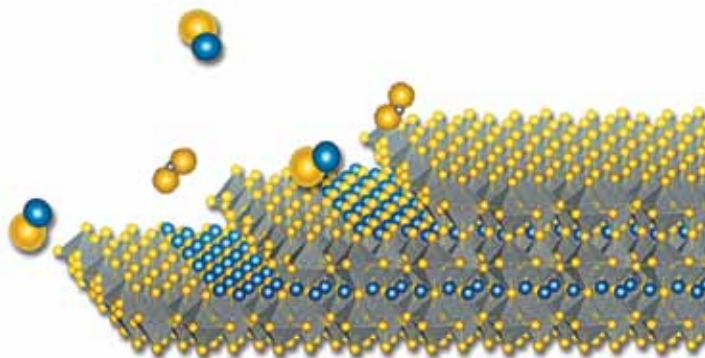


Spacecrafts often encounter a continuous stream of damaging cosmic radiation, which can harm or even destroy onboard electronics. To extend future missions, researchers reporting in *ACS Nano* show that transistors and circuits with carbon nanotubes can be configured to maintain their electrical properties and memory after being bombarded by high amounts of radiation. To do this, carbon nanotubes were deposited on a silicon wafer as the semiconducting layer in field-effect transistors, and then tested for different transistor configurations with various levels of shielding around the semiconducting layer. It was found that placing shields both above and below the carbon nanotubes protected the transistor's electrical properties against incoming radiation up to 10 Mrad—a level much higher than most silicon based radiation-tolerant electronics can handle.

A memory chip was made of transistors with carbon nanotubes that maintained their electrical properties and memory after being bombarded by high amounts of radiation (Credit: [www.acs.org/content](http://www.acs.org/content))

## Ultra-stable semiconductors made of inexpensive materials

MIT engineers have reportedly created the first high-quality thin films of a new family of semiconductor materials. The feat has the potential to impact multiple fields of technology for computers, solar cells, night-vision cameras, and more. Known as chalcogenide perovskites, the new family of semiconductors is ultra-stable and made of inexpensive, nontoxic elements. The thin films are composed of barium, zirconium, and sulphur in a specific crystal structure. Several variations can be made by simply varying the composition. A technique called molecular beam epitaxy was used to grow high-quality films during the testing process.



Artist's conception of the epitaxial growth of a chalcogenide perovskite thin film. The material represents a new family of semiconductors (Credit: <https://news.mit.edu>)